WORKED SOLUTION

STEP I 2019, Question 1

Pure •

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Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Everything follows from where the line cuts the axes. For the perimeter, the fixed point $P = (1, k)$ splits the hypotenuse XY into two pieces, each the hypotenuse of a small right-angled triangle with a side of known length, so no square roots need to be expanded. For the minima, the area is a sum of two terms with a constant product (an AM–GM situation), and the derivative of the perimeter collapses beautifully once we use $\cos^2\theta = (1 - \sin\theta)(1 + \sin\theta)$.

Solution

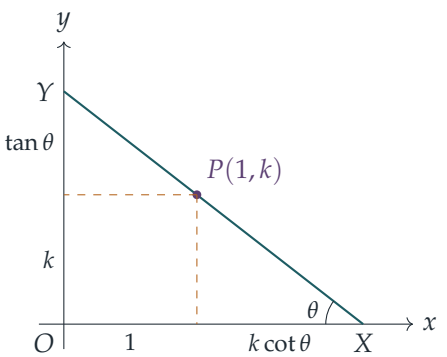
The line through $(1, k)$ with gradient $-\tan\theta$ is

$$y - k = -\tan\theta\,(x - 1).$$

Putting $y = 0$ gives $x - 1 = k\cot\theta$, and putting $x = 0$ gives $y = k + \tan\theta$, so

$$X = (1 + k\cot\theta,\ 0), \qquad Y = (0,\ k + \tan\theta).$$

Since $k > 0$ and $0 < \theta < \frac{1}{2}\pi$, both intercepts are positive, so OXY is a genuine right-angled triangle in the first quadrant, with $P = (1, k)$ on its hypotenuse.



Part (i)

The triangle has a right angle at O, so

$$\begin{aligned} A = \tfrac{1}{2}(1 + k\cot\theta)(k + \tan\theta) &= \tfrac{1}{2}\big(k + \tan\theta + k^2\cot\theta + k\big) \\ &= k + \tfrac{1}{2}\big(\tan\theta + k^2\cot\theta\big), \end{aligned}$$

using $\cot\theta\tan\theta = 1$.

For the minimum, both $\tan\theta$ and $k^2\cot\theta$ are positive and their product is the constant k^2. By the AM–GM inequality,

$$\tan\theta + k^2\cot\theta \geq 2\sqrt{\tan\theta \cdot k^2\cot\theta} = 2k,$$

with equality exactly when $\tan\theta = k^2\cot\theta$, that is, $\tan\theta = k$ (which is attained by some θ in the range, since $k > 0$). Hence $A \geq k + k = 2k$, and

$$A_{\min} = 2k.$$

(Differentiating gives the same: $\frac{\mathrm{d}A}{\mathrm{d}\theta} = \frac{1}{2}(\sec^2\theta - k^2\operatorname{cosec}^2\theta) = 0$ when $\tan^2\theta = k^2$.) At the minimum $X = (2, 0)$ and $Y = (0, 2k)$, so P is the midpoint of XY.

Part (ii)

The two legs are $OX = 1 + k\cot\theta$ and $OY = k + \tan\theta$. For the hypotenuse, split it at P as in the diagram.

- From P to Y: the horizontal run is 1 and the line makes angle θ with the horizontal, so $PY = 1/\cos\theta = \sec\theta$.
- From P to X: the vertical drop is k, so $PX = k/\sin\theta = k\operatorname{cosec}\theta$.

Adding the four lengths,

$$L = 1 + \tan\theta + \sec\theta + k\big(1 + \cot\theta + \operatorname{cosec}\theta\big),$$

as required.

> **Ask yourself** The derivative contains $\sec^2\theta + \sec\theta\tan\theta$. Is there a way to make it much simpler before setting it to zero?

Yes. Differentiating term by term,

$$\frac{\mathrm{d}L}{\mathrm{d}\theta} = \sec^2\theta + \sec\theta\tan\theta - k\big(\operatorname{cosec}^2\theta + \operatorname{cosec}\theta\cot\theta\big) = \frac{1 + \sin\theta}{\cos^2\theta} - k\,\frac{1 + \cos\theta}{\sin^2\theta}.$$

Now $\cos^2\theta = (1 - \sin\theta)(1 + \sin\theta)$ and $\sin^2\theta = (1 - \cos\theta)(1 + \cos\theta)$, and $1 + \sin\theta$, $1 + \cos\theta$ are non-zero, so we may cancel:

$$\frac{\mathrm{d}L}{\mathrm{d}\theta} = \frac{1}{1 - \sin\theta} - \frac{k}{1 - \cos\theta}.$$

This is zero exactly when $1-\cos\theta = k(1-\sin\theta)$. So the minimum occurs at $\theta = \alpha$ with

$$\frac{1-\cos\alpha}{1-\sin\alpha} = k,$$

which is what we had to show. (In fact this stationary point is the only one and is a minimum: as θ increases on $(0, \frac{1}{2}\pi)$, the first term of $\frac{\mathrm{d}L}{\mathrm{d}\theta}$ increases and the second decreases, so $\frac{\mathrm{d}L}{\mathrm{d}\theta}$ is increasing and changes sign from negative to positive once.)

Ask yourself How do we get rid of k in the minimum value?

Group the terms of L over common denominators:

$$1+\tan\alpha+\sec\alpha = \frac{1+\sin\alpha+\cos\alpha}{\cos\alpha}, \qquad 1+\cot\alpha+\operatorname{cosec}\alpha = \frac{1+\sin\alpha+\cos\alpha}{\sin\alpha}.$$

Writing $s = \sin\alpha$, $c = \cos\alpha$ and $k = \dfrac{1-c}{1-s}$, the second piece becomes

$$k\,\frac{1+s+c}{s} = \frac{(1+s+c)(1-c)}{s(1-s)} = \frac{1+s-sc-c^2}{s(1-s)} = \frac{s^2+s-sc}{s(1-s)} = \frac{1+s-c}{1-s},$$

using $1-c^2 = s^2$. Therefore

$$\begin{aligned} L_{\min} = \frac{1+s+c}{c} + \frac{1+s-c}{1-s} &= \frac{(1+s+c)(1-s)+c(1+s-c)}{c(1-s)} \\ &= \frac{1-s^2+c-sc+c+sc-c^2}{c(1-s)} \\ &= \frac{2c}{c(1-s)}, \end{aligned}$$

since $1-s^2-c^2 = 0$. So

$$L_{\min} = \frac{2}{1-\sin\alpha}.$$

Checks

Take $k = 1$. Then $\tan\theta = 1$ gives $X = (2,0)$, $Y = (0,2)$ and $A = 2 = 2k$, as predicted. The perimeter condition $(1-\cos\alpha)/(1-\sin\alpha) = 1$ gives $\alpha = \frac{1}{4}\pi$, and $2/(1-\frac{1}{\sqrt{2}}) = 4+2\sqrt{2} \approx 6.828$, which is exactly $2+2+2\sqrt{2}$ for the triangle with vertices $(0,0)$, $(2,0)$, $(0,2)$. For a range of other values of k, a computer search over θ confirmed both minimum values and the location of the perimeter minimum.

Common slips

- Expanding $\sqrt{(1+k\cot\theta)^2+(k+\tan\theta)^2}$ for the hypotenuse instead of splitting it at P.

- Setting the unsimplified derivative to zero and getting lost; cancel $1+\sin\theta$ and $1+\cos\theta$ first.
- Leaving k in the final answer: the question wants the minimum in terms of α alone.

Beyond STEP

The neat answer has a geometric explanation. The minimum-perimeter line through P is the one tangent at P to a circle touching both axes (an excircle of OXY), and the perimeter is then twice the tangent length from O to that circle; this is the circle of radius r through P with $r = 1/(1-\sin\alpha)$. Optimisation problems with a constraint like "the line passes through P" are the starting point for Lagrange multipliers in a first-year calculus course. For many more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.